

Lekai Qian

📍 Guangzhou, China | ✉ lekaiqian@outlook.com | 🌐 Lekai-Qian.github.io

Education

South China University of Technology, B.E. in Artificial Intelligence (*Expected*) 2023.09 - *Expected* 2027.06

- GPA: 3.84/4.0
- **Relevant Coursework:** Machine Learning, Natural Language Processing, Data Structures, Digital Signal Processing, Probability Theory, Linear Algebra

Research Experience

BEAT: Tokenizing and Generating Symbolic Music by Uniform Temporal Steps

First author, ICML 2026 [arXiv] [Demo]

Supervisor: Prof. Gus Xia, Dr. Ziyu Wang, MBZUAI Music X Lab

- Proposed BEAT, a beat-level tokenization that uses uniform musical time steps as causal modeling units, enabling one autoregressive Transformer framework for continuation, pattern-controlled generation, and real-time accompaniment; achieved the best JSGC/FMD on piano and multi-track continuation, with lower BPE compression rate (64.83% vs. 80.15%/80.18% at 20 merges).

Real-Time Language Model Jamming: A Case Study for Live Music Accompaniment Generation

RTAS 2026 [arXiv]

Supervisor: Prof. Xiaosong Ma, MBZUAI Music X Lab

- Adapted the BEAT-based generative modeling algorithm to StreamMUSE's real-time accompaniment system, integrating beat-synchronized conditioning with the streaming pipeline; achieved $ISR = 0.9536$ and $ISRw = 0.9104$ in local-server deployment, and maintained functional $ISRw = 0.8785$ under remote-server conditions.

Pianoroll-Event: A Novel Score Representation for Symbolic Music

First author, ICASSP 2026 [arXiv] [Demo]

- Proposed Pianoroll-Event, an early grid-to-sequence representation that converts sparse pianoroll frames into compact token sequences, improving encoding efficiency by $1.36\times$ to $7.16\times$ over common symbolic music tokenizations.

Anchored Cyclic Generation for Long-Sequence Symbolic Music

Findings of ACL 2026

- Proposed a global-to-local anchored generation framework for long-sequence symbolic music, reducing semantic-vector cosine distance by 34.7%.

Diffusion-Based Symbolic Music Generation (University SRP Project)

Team Leader

- Led a team project on cascaded diffusion models for symbolic music generation, covering dataset curation, model design, and training.

Technologies

Deep Learning: Generative Modeling, Transformers, Sequence Modeling, Diffusion Models, Distributed Training

Programming & Tools: Python, PyTorch, Hugging Face, Git/GitHub

Domain Expertise: Music Information Retrieval, Symbolic Music Generation, Representation Learning, Audio Signal Processing